

Simple Linear Regression

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Introduction

Simple linear regression models the mean of a continuous outcome as a linear function of a single continuous predictor. It is the workhorse of applied statistics: half of all reported analyses are linear regressions in some disguise, and the concepts it introduces (least squares estimation, residuals, fit-vs-data trade-off, R-squared, confidence and prediction bands) generalise to every more complex model.

Prerequisites

The reader should know what a mean, a variance, and a correlation are, and should be comfortable reading a scatter plot. No prior regression experience is required.

Theory

Given pairs (x_i, y_i) for $i = 1, \dots, n$, the simple linear regression model is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2).$$

The parameters β_0 (intercept) and β_1 (slope) are estimated by ordinary least squares – minimising the sum of squared residuals $\sum (y_i - \hat{y}_i)^2$. Closed-form solutions are

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

The standard error of the slope is $SE(\hat{\beta}_1) = s / \sqrt{\sum (x_i - \bar{x})^2}$, where s is the residual standard error. Under the normality assumption, $\hat{\beta}_1$ is normally distributed around β_1 , and $\hat{\beta}_1 / SE(\hat{\beta}_1)$ follows a t distribution with $n - 2$ degrees of freedom under the null $\beta_1 = 0$.

The coefficient of determination is $R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$ – the proportion of variance in y explained by the linear relationship with x . For simple regression, R^2 equals the squared Pearson correlation.

Assumptions

The OLS inference machinery relies on four assumptions:

1. **Linearity**: the true relationship between x and the mean of y is linear.
2. **Independence**: the errors ε_i are independent of each other.
3. **Homoscedasticity**: the errors have constant variance σ^2 regardless of x .
4. **Normality of errors**: the errors are normally distributed – important for small-sample inference, less so for large samples.

R Implementation

```

library(broom)
library(ggplot2)
library(performance)

set.seed(7)
n <- 80
x <- runif(n, 0, 10)
y <- 2 + 1.5 * x + rnorm(n, sd = 1.5)
df <- data.frame(x, y)

fit <- lm(y ~ x, data = df)
tidy(fit, conf.int = TRUE)
glance(fit)

ggplot(df, aes(x = x, y = y)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", colour = "steelblue") +
  theme_minimal(base_size = 12)

check_model(fit)

```

The `tidy()` function returns a one-row-per-term table with estimates, standard errors, t statistics, p-values, and confidence intervals. `glance()` returns a one-row model-level summary: R-squared, adjusted R-squared, residual standard error, F statistic, and log-likelihood. `check_model()` produces the standard diagnostic panel.

Output & Results

Typical coefficient output:

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	1.982	0.326	6.080	<0.001	1.333	2.631
x	1.512	0.056	27.155	<0.001	1.401	1.623

$R^2 = 0.904$, residual SE = 1.47, $F(1, 78) = 737.4$.

Interpretation

The slope estimate of 1.512 (95% CI 1.401 to 1.623) says that a one-unit increase in x is associated with a 1.51-unit increase in the conditional mean of y , holding all else equal (vacuous for a simple regression, but the phrasing generalises). The intercept of 1.982 is the predicted y when $x = 0$; this is only a scientifically meaningful quantity if $x = 0$ is within the range of the data.

For a manuscript: “A simple linear regression showed that y increased significantly with x , $b = 1.51$ (95% CI 1.40 to 1.62), $t(78) = 27.2$, $p < 0.001$. The model explained 90.4% of the variance in y .”

Practical Tips

- Always plot the data and the fitted line; numbers alone can hide gross misfit.
- Residuals-vs-fitted and Q-Q plots are the two diagnostics you should reflexively inspect.
- An R^2 of 0.9 is stellar in physics, mediocre in psychology, and suspect in epidemiology – interpret it in context.

- The confidence interval is for the *mean* response at a given x ; the prediction interval, which is wider, is for a *new individual* observation.
- Extrapolating beyond the range of the observed x is risky: the linear relationship that held for $x \in [0, 10]$ may be nonsense for $x = 100$.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI